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SOURCE: MSC.36(63).pdf
not endanger the craft, other machinery, occupants of the craft or any other persons.
Requirements of 9.2.6 should apply to gas turbines in respect of fuel which might reach the interior of the jet pipe or exhaust system after a false start or after stopping.
Turbines should be safeguarded as far as practicable against the possibility of damage by ingestion of contaminants from the operating environment. Information regarding the recommended maximum concentration of contamination should be made available.
Provision should be made for preventing the accumUlation of salt deposits on the compressors and turbines and, if necessary, for preventing the air intake from icing.
In the event of a failure of a shaft or weak link, the ~roken end should not hazard the occupants of the craft, either directly or by damaging the craft or its systems. Where necessary, guards may be fitted to achieve compliance with these requirements.
Each engine should be provided with an emergency overspeed shutdown device connected, where possible, directly to each rotor shaft.
Where an acoustic enclosure is fitted which completely surrounds the

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SOURCE: MSC.36(63).pdf
Each engine should be provided with an emergency overspeed shutdown device connected, where possible, directly to each rotor shaft.
Where an acoustic enclosure is fitted which completely surrounds the gas generator and the high pressure oil pipes, a fire detection and extinguishing system should be provided for the acoustic enclosure.
Details of the manufacturers' proposed automatic safety devices to guard against hazardous conditions arising in the event of malfunction in the turbine installation should be provided together with the failure mode and effect analysis.
The manufacturers should demonstrate the soundness of the casings. Intercoolers and heat exchangers should be hydraulically tested on each side separately.
Diesel engines for main propulsion and essential auxiliaries
Any main diesel propulsion system should have satisfactory torsional vibration and other vibrational characteristics verified by individual and combined torsional and other vibration analyses for the system and its components from power unit through to propulsor.
W/1687n/pp/EWP
RESOLUTION MSC.36(63)
(adopted on 20 May 1994)
ADOPTION OF THE INTERNATIONAL CODE OF SAFETY

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SOURCE: MSC.36(63).pdf
individual and combined torsional and other vibration analyses for the system and its components from power unit through to propulsor.
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9.4.4
9.4.5
9.4.6
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9.5.1
9.5.2
9.5.3
All external high pressure fuel delivery lines between the high pressure fuel pwnps and fuel nozzles should be protected with a jacketed tubing system capable of containing fuel from a high pressure line failure. The jacketed tubing system should include a means for collection of leakages and arrangements should be provided for an alarm to be given of a fuel line failure.
Engines of a cylinder diameter of 200 mm or a crankcase volwne of 0.6 m3 and above should be provided with crankcase explosion relief valves of an approved type with sufficient relief area. The relief valves should be arranged with means to ensure that discharge from them is directed so as to minimize the possibility of injury to personnel.
The lubrication system and arrangements should be efficient at all

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SOURCE: MSC.36(63).pdf
relief valves should be arranged with means to ensure that discharge from them is directed so as to minimize the possibility of injury to personnel.
The lubrication system and arrangements should be efficient at all running speeds, due consideration being given to the need to maintain suction and avoid the spillage of oil in all conditions of list and trim and degree of motion of the craft.
Arrangements should be provided to ensure that visual and audible alarms are activated in the event of either lubricating oil pressure or lubricating oil level falling below a safe level, considering the rate of circulation of oil in the engine. Such events should also cause automatic reduction of engine speed to a safe level, but automatic shutdown should only be activated by conditions leading to a complete breakdown, fire or explosion.
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Where diesel engines are arranged to be started, reversed or controlled by compressed air, the arrangement of the air compressor, air receiver and air starting system should be such as to minimize the risk of fire or explosion.
Transmissions
The transmission should be of adequate strength and stiffness to

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SOURCE: MSC.36(63).pdf
air receiver and air starting system should be such as to minimize the risk of fire or explosion.

Transmissions

The transmission should be of adequate strength and stiffness to enable it to withstand the most adverse combination of the loads expected in service without exceeding acceptable stress levels for the material concerned.

The design of shafting, bearings and mounts should be such that hazardous whirling and excessive vibration could not occur at any speed up to 105% of the shaft speed attained at the designed overspeed trip setting of the prime mover.

The strength and fabrication of the transmission should be such that the probability of hazardous fatigue failure under the action of the repeated loads of variable magnitude expected in service is extremely remote throughout its operational life. Compliance should be demonstrated by suitably conducted tests, and by designing for sufficiently low stress levels, combined with the use of fatigue resistant materials and suitable detail design. Torsional vibration or oscillation likely to cause failure may be acceptable if it occurs at transmission speeds which would not be used in normal

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SOURCE: MSC.36(63).pdf
resistant materials and suitable detail design. Torsional vibration or oscillation likely to cause failure may be acceptable if it occurs at transmission speeds which would not be used in normal craft operation, and it is recorded in the craft operating manual as a limitation.

Where a clutch is fitted in the transmission, normal engagement of the clutch should not cause excessive stresses in the transmission or driven items. Inadvertent operation of any clutch should not produce dangerously high stresses in the transmission or driven item.

Provision should be made such that a failure in any part of the transmission, or of a driven component, will not cause damage which might hazard the craft or its occupants.

Where failure of lubricating fluid supply or loss of lubricating fluid pressure could lead to hazardous conditions, provision should be made to enable such failure to be indicated to the operating crew

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SOURCE: MSC.36(63).pdf
might hazard the craft or its occupants.
Where failure of lubricating fluid supply or loss of lubricating fluid pressure could lead to hazardous conditions, provision should be made to enable such failure to be indicated to the operating crew in adequate time to enable them as far as practicable to take the appropriate action before the hazardous condition arises.
Propulsion and lift devices
The requirements of this section are based on the premise that:
. 1 Propulsion arrangements and lift arrangements may be provided by separate devices, or be integrated into a single propulsion and lift devices. Propulsion devices may be air, or water propellers or water jets and the requirements apply to all types of craft.
.2 Propulsion devices are those which directly provide the propulsive thrust and include machinery items and any , . associated ducts, vanes, scoops and nozzles, the prlmary function of which is to contribute to the propulsive thrust .